Discrete Mathematics
Tutorial 01.

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03. a) P → It is your birthday
Q → There will be cake.
$(P \vee Q) \rightarrow Q$

b)

P	Q	P ∨ Q	(P ∨ Q) → Q
T	T	T	T
T	F	T	F
F	T	T	T
F	F	F	T

c) only that there will be cake.

d) Not your birthday

e) It is your birthday, but cake is lie.

04. a) P : number is triangular
Q : number is square
$\neg R$: It is not prime.

$(P \vee Q) \rightarrow \neg R$

p	q	R	~R	p v q	(p v q) → ~R
1	1	1	0	1	0
1	1	0	1	1	1
1	0	1	0	1	0
1	0	0	1	1	1
0	1	1	0	1	0
0	1	0	1	1	1
0	0	1	0	0	1
0	0	0	1	0	0

b) when R is false in the above truth table.

p	q	R	(p v q)	~R	(p v q) → ~R
T	T	T	T	F	F
T	F	T	T	F	F
F	T	T	T	F	F

∴ The statement is false, then it is required to find the numbers with the last column of the table with false.

c)

p	q	R	p v q	(p v q) → R
1	1	1	1	1
0	1	1	1	1
1	0	1	1	1
0	0	1	0	1
1	1	0	1	0
0	1	0	1	0
1	0	0	1	0
0	0	0	0	1

[03]

» If the statement were true and the number 5867 is definitely prime, the number is not triangular and not a square number.

05. I $\neg(P \to Q)$ and $(P \wedge \neg Q)$

This is logically equivalent by logical to

P	Q	$P \to Q$	$\neg Q$	$\neg(P \to Q)$	$(P \wedge \neg Q)$
1	1	1	0	0	0
1	0	0	1	1	1
0	1	1	0	0	0
0	0	1	1	0	0

I $P \to (Q \vee R) \equiv (P \to Q) \vee (P \to R)$

This is logically equwalent.

P	Q	R	$Q \vee R$	$P \to (Q \vee R)$	$P \to Q$	$P \to R$	$(P \to Q) \vee (P \to R)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1
0	1	0	1	1	1	1	1
0	0	1	1	1	1	1	1
0	0	0	0	1	1	1	1

$P \to (Q \vee R) \equiv (P \to Q) \vee (P \to R)$

06. a) For all integers a and b
if a or b is odd
∴ a+b is odd

b) for all integers a and b
if a and b are even
∴ a+b is even

c) There are numbers a and b such that a+b is even
~~a and b not both are~~
a and b not both are even.

d) false
~~Eg:-~~
eg:-
a = 1 ; b = 1.
a+b = 2, which is even.
but both a and b are odd.
So the original statement is false.

e) false, Since it is equivalent to the original statement.

f) True.
Assume a and b are even.
a = 2S and b = 2R
then a+b ⇒ 2S + 2R
2(S+R) which is even.

g) True.
Since the statement is false.

07. The game TENZI comes with 40 six-sided dice (each number 1-6). Suppose you roll all 40 dice.

a) Suppose that each number only came up 06 or, fewer times. so there are at most six 1's, six 2's and go on. That's a totall of 36 dice. So you must not have rolled all 40 dices.

b) we can have a dice without any four matching or, any four being all different. There is, three 1's, three 2's. we will prove that whenever you roll 10 dice, you will always get four matching or, all being different.

Suppose you roll 10 dice, But that there are not four matching rolls. This means at most there are three of any given value. If we only had three different values, that would be only a dice, So there must be 4 different values, giving a dice that are all different.

11. First digit of the local number.

* It can not zero. So there are 9 possibilities (1-9)

Remaining six digits of the local numbers.

* It's 10 possibilities (0-9)

Total numbers of possible.

9×10^6

$\Rightarrow 9\,000\,000$.

12. 1, 2, 3, 5

a) $\frac{1}{1}$ $\frac{1,2,3,5}{4}$ $\frac{1,2,3,5}{4}$ $\frac{1,2,3,5}{4}$

There is one possible 1st digit and 4 second, third and fourth

$1 \times 4 \times 4 \times 4 \Rightarrow 64$.

b) $\frac{1}{1}$ $\frac{2,3,5}{3}$ $\frac{3,5}{2}$ $\frac{5}{1}$

one possible for digit, 3 for second, 2 for 3rd and again one for last

$1 \times 3 \times 2 \times 1 = 6$.

c) $\frac{1,2,3,5}{4}$ $\frac{1,2,3,5}{4}$ $\frac{1,2,3,5}{4}$ $\frac{2}{1}$

four possible digits for first, second and third. 1 digit for fourth

$4 \times 4 \times 4 \times 1 = 64$.

d) $\frac{1,3,5}{3}$ $\frac{3,5}{2}$ $\frac{5}{1}$ $\frac{2}{1}$

03 possible for 1st, 2 for second and one for third and fourth

$3 \times 2 \times 1 \times 1 = 6$.

10. a) $\binom{10}{3} \Rightarrow \frac{10 \times 9 \times 8}{3 \times 2 \times 1} \rightarrow 120$

b) $2^{10} \rightarrow 1024$ pizzas.

c) $P(10,5) = 10 \times 9 \times 8 \times 7 \times 6$

$\rightarrow 30240$.

21. 2 through 8 (2, 3, 4, 5, 6, 7, 8)

a) multiplicative principle

$7 \times 7 \times 7 \times 7 \times 7$

$7^5 = 16807$ numbers.

b) $7 \times 6 \times 5 \times 4 \times 3$

$= 2520$ numbers.

c) 7, 6, 5, 4, 3

$\binom{7}{5} = \frac{7 \times 6 \times 5 \times 4 \times 3}{5 \times 4 \times 3 \times 2 \times 1} = 21$ numbers.

d) The premutation is in part (b), while the combination is in part (c), At first this seems backwards. Since usually we use combinations for when order does not matter. here it looks like in part (c) that order does matter. The better way to distinguish between combinations and premutations is to ask whether we are counting different arrangements as different outcomes. In part (c) there is only one arrangement of any set of 5 digits, while in part (b) each set of 5 digits gives, Then we can get different outcomes.

55.



a) $\binom{7}{2}\binom{7}{2} = \left[\frac{7\times6}{2\times1}\right] \times \left[\frac{7\times6}{2\times1}\right] = 21\times21 = 441$,

b) There are 21 Squares.

c) $\left[\frac{7}{2}\right] \times \left[\frac{7\times6}{2\times1}\right] = 21$ Rectangle.

d) $\left[\frac{7}{2}\right] + \left[\frac{7-1}{2}\right] + \left[\frac{7-2}{2}\right] + \left[\frac{7-4}{2}\right] + \left[\frac{7-6}{2}\right] + \left[\frac{7-8}{2}\right]$

$= 21 + 20 + 18 + 15 + 11 + 6$

$\Rightarrow 91$

e) All of the quadrilaterals - Trapezoids.

So, no. of Trapezoids $= {}^7C_2 \times {}^7C_2$

$\Rightarrow 21 \times 21 \Rightarrow 441$

f) $\left[\binom{7}{2}\binom{7}{2}\right] - \left[\left[\frac{7}{2}\right] + \left[\frac{7-1}{2}\right] + \left[\frac{7-3}{2}\right] + \left[\frac{7-4}{2}\right] + \left[\frac{7-6}{2}\right] + \left[\frac{7-8}{2}\right]\right]$

$\Rightarrow 441 - 91$

$\Rightarrow 350$.